

Processamento de Imagens

Filtros e Convolução

Professora Leo Sampaio Ferraz Ribeiro

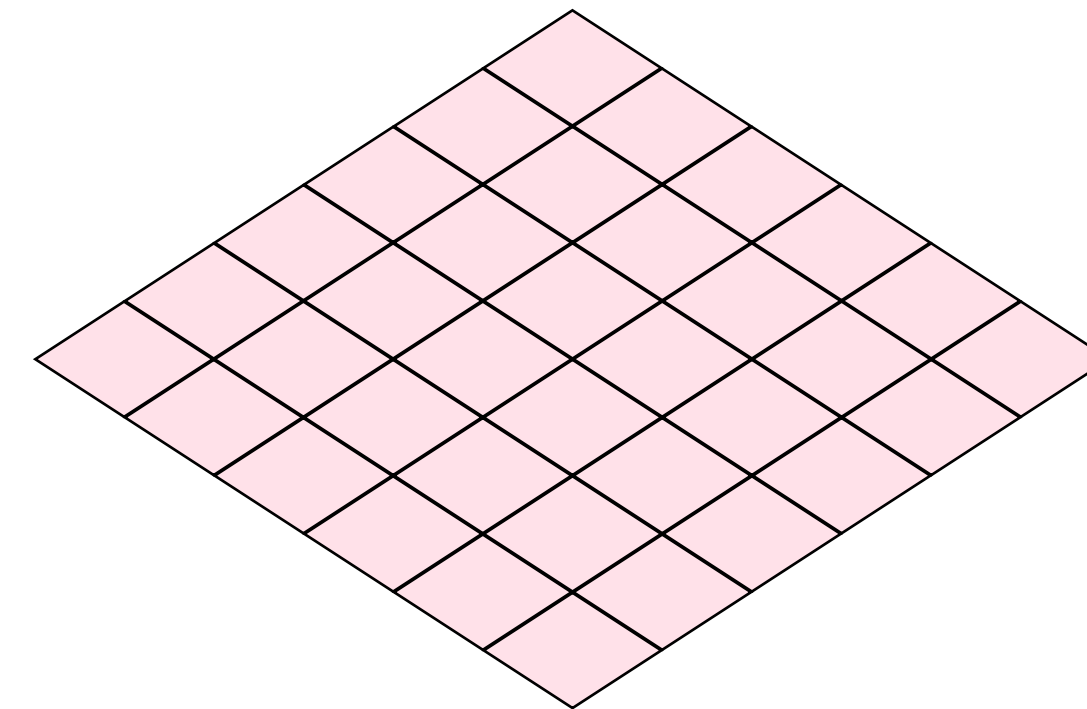
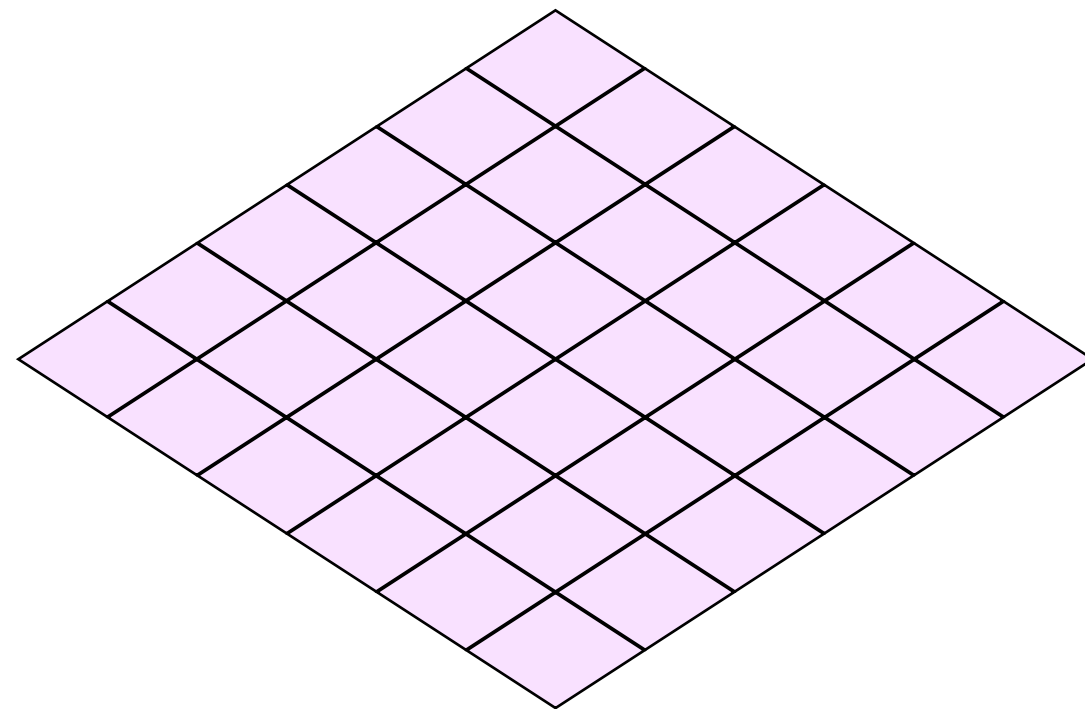


Slide para não esquecer de passar a lista

[illegible]

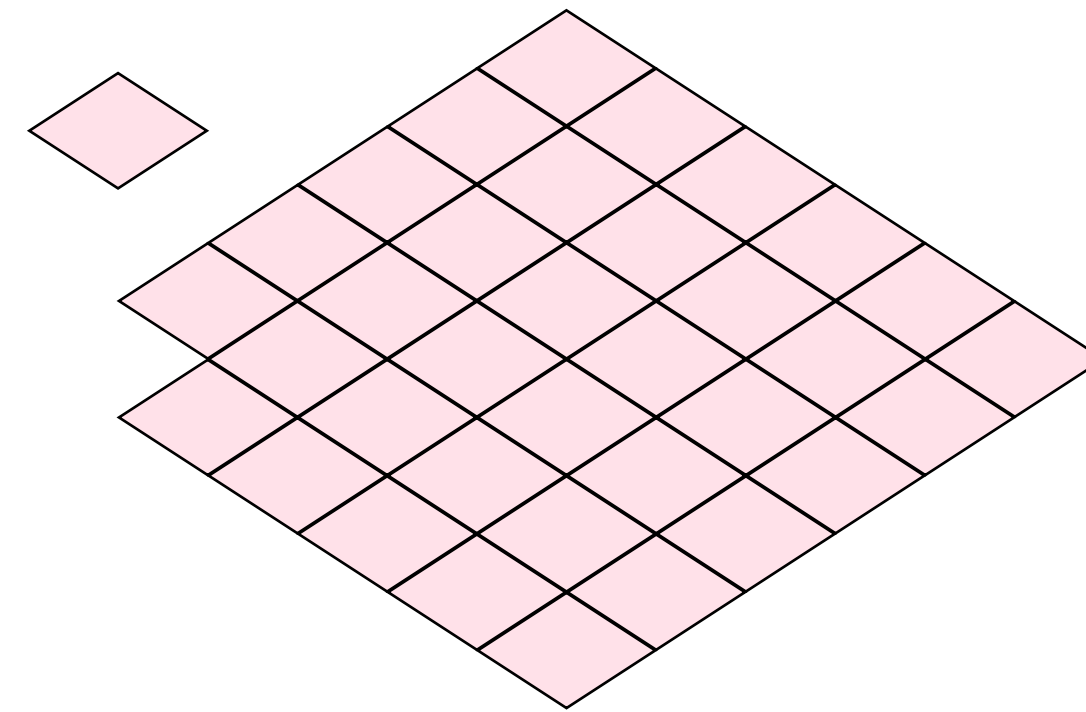
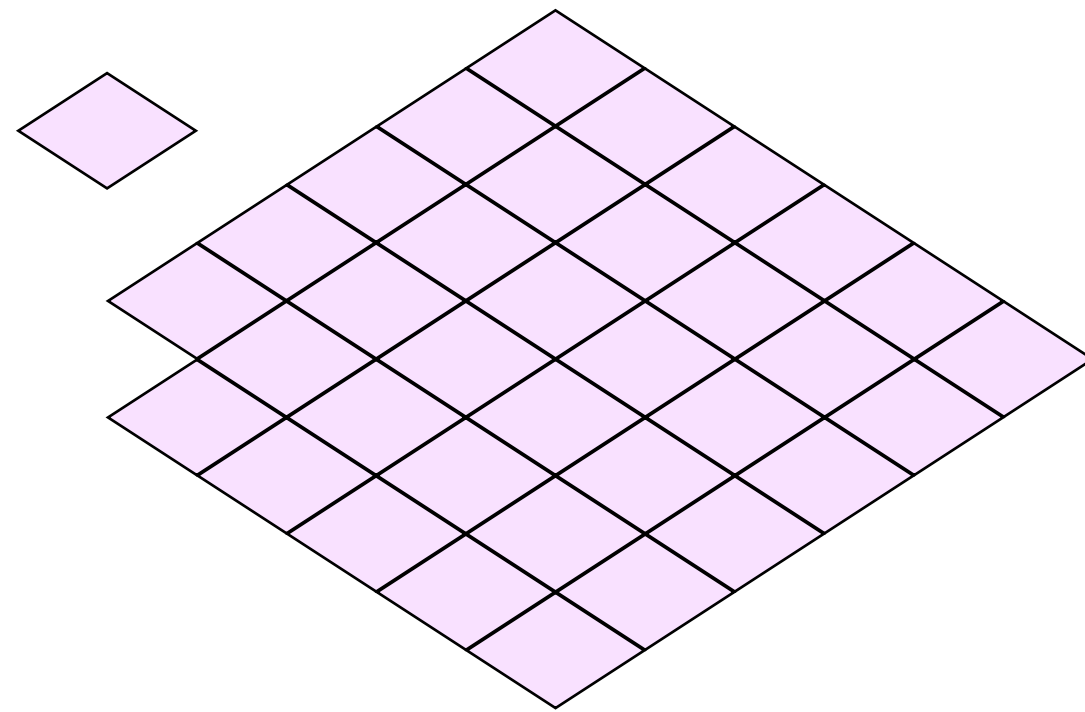
Comparação

Até agora estávamos olhando transformações de um ponto para outro ponto



Comparação

Até agora estávamos olhando transformações de um ponto para outro ponto

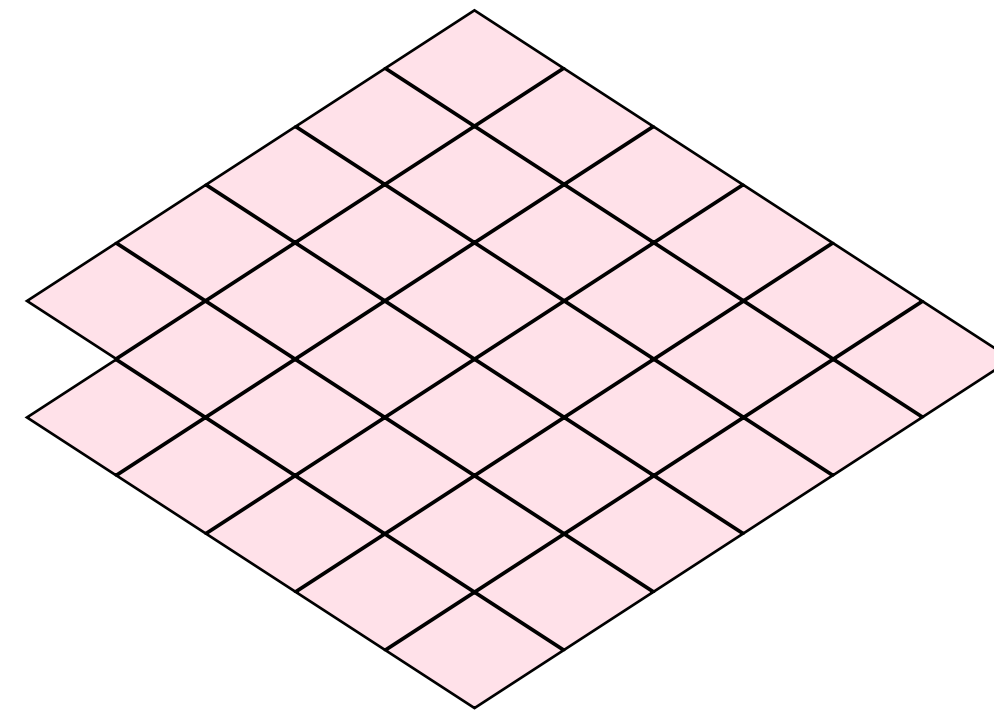
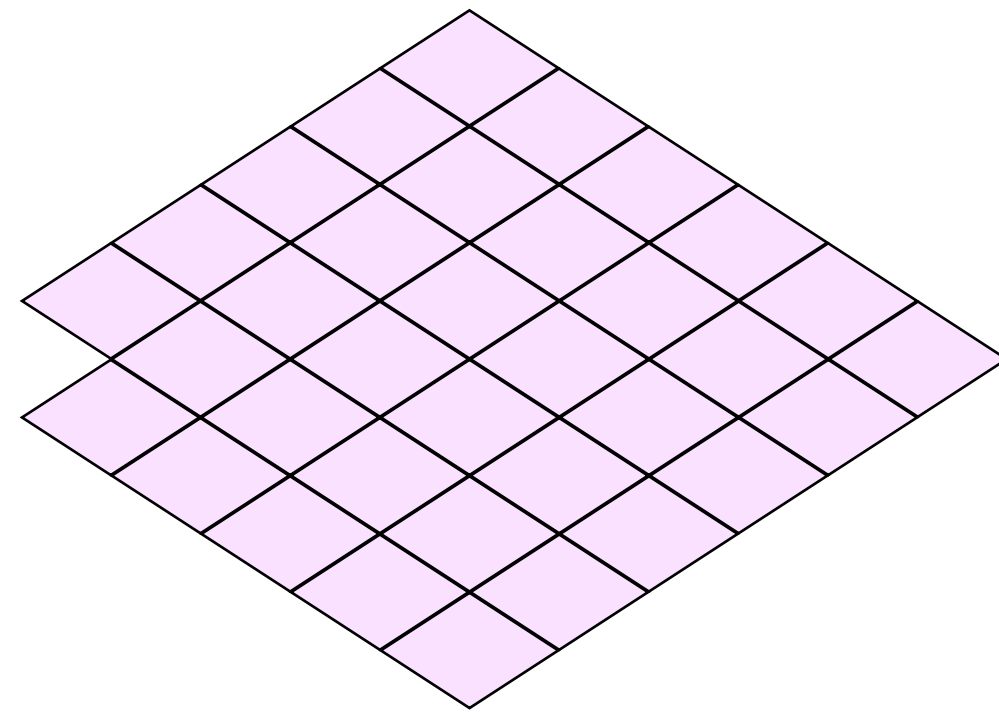


$$T(z)$$

Comparação

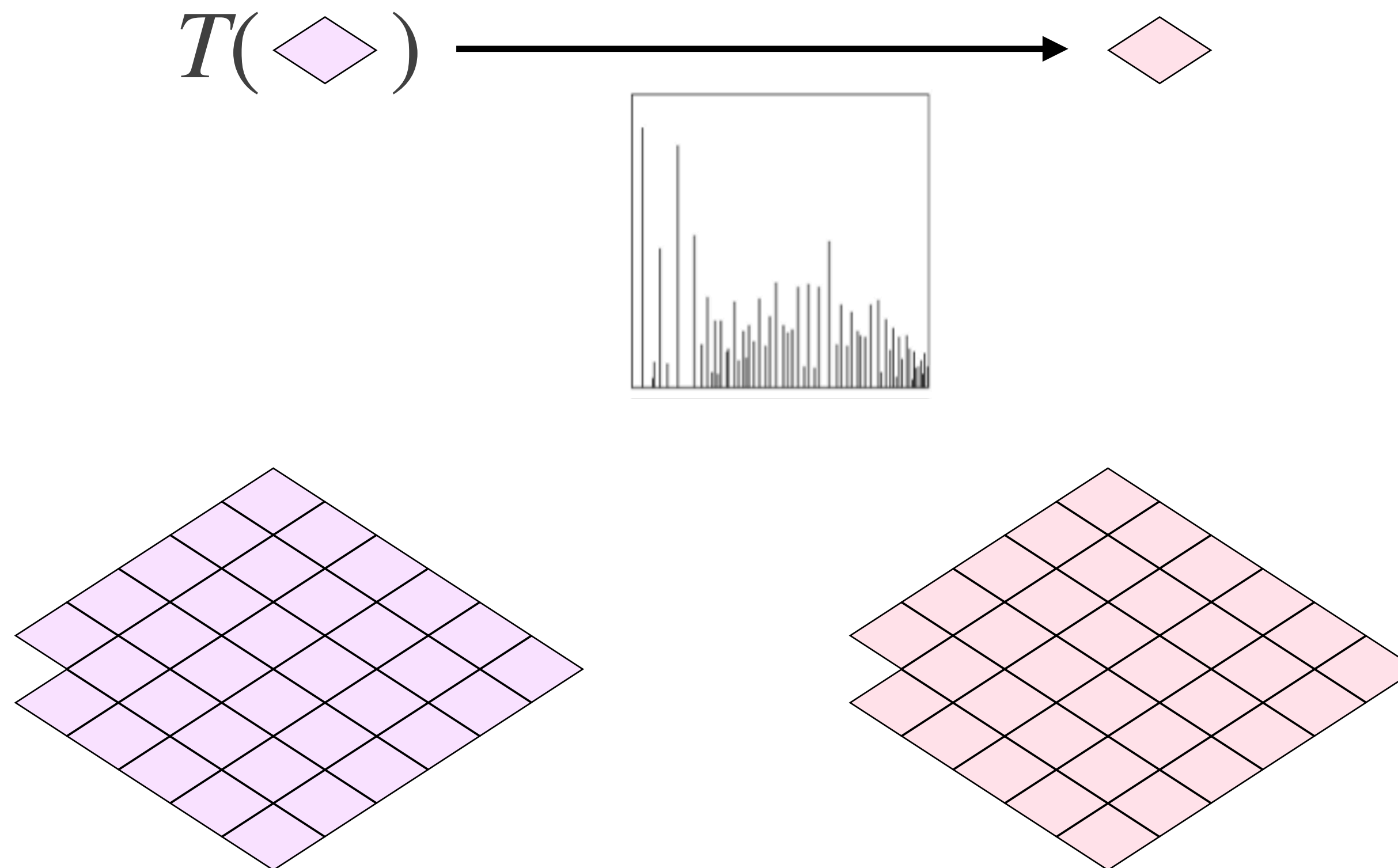
Até agora estávamos olhando transformações de um ponto para outro ponto

$$T(\text{◇}) \longrightarrow \text{◇}$$



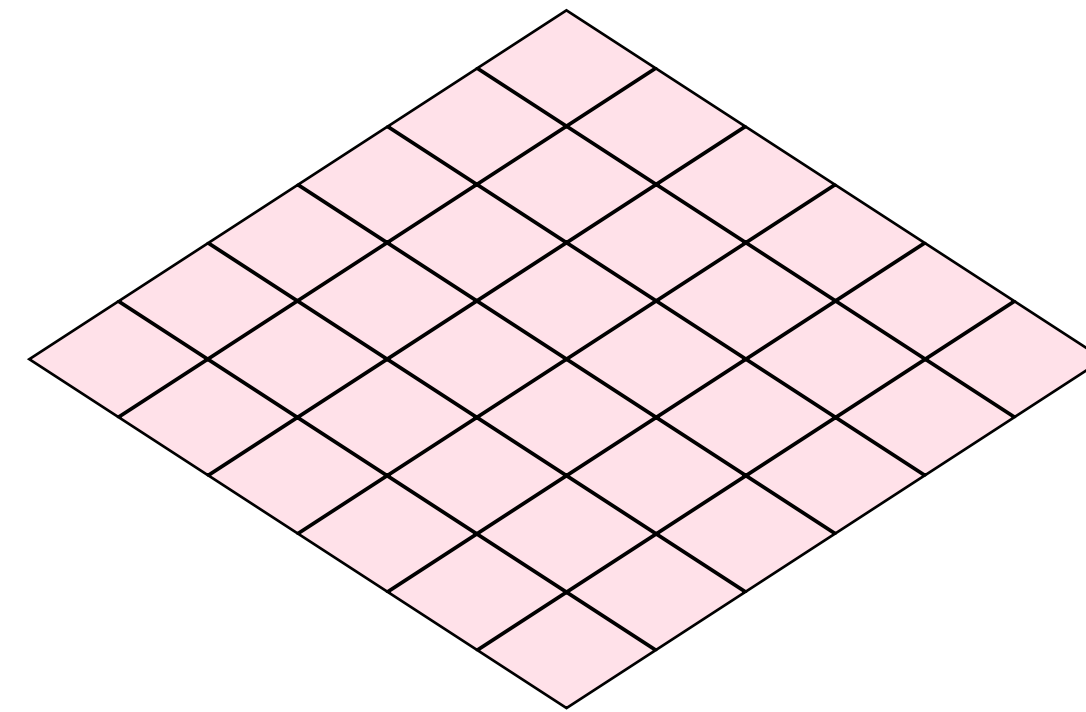
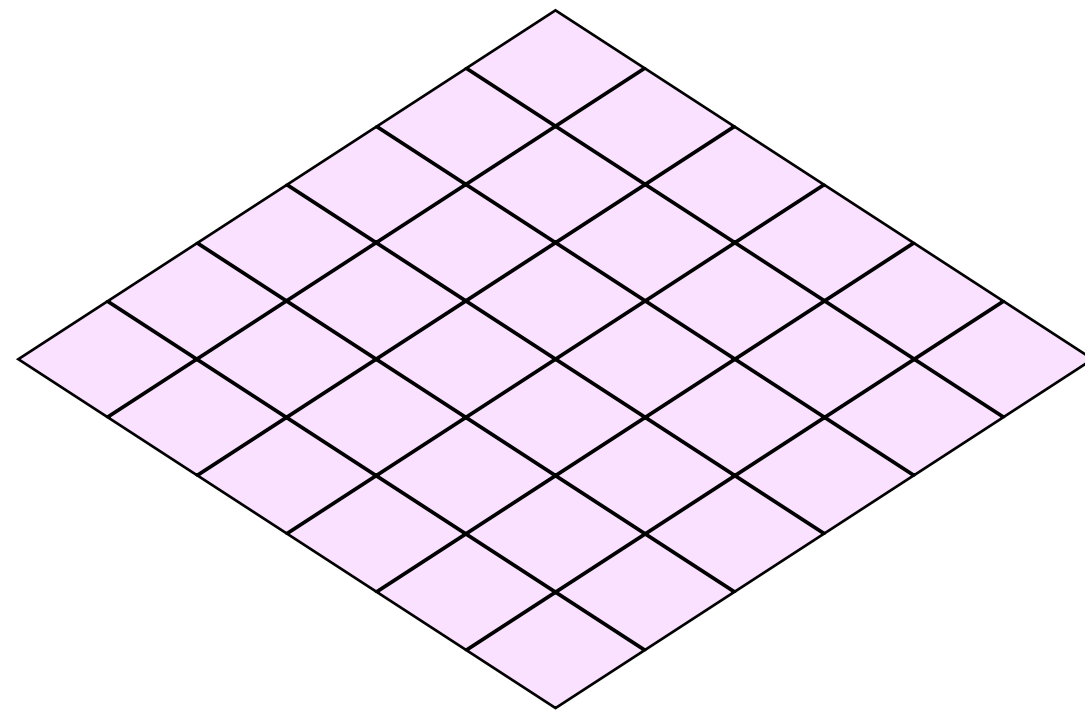
Comparação

Mesmo no caso em que usamos o histograma para determinar a função



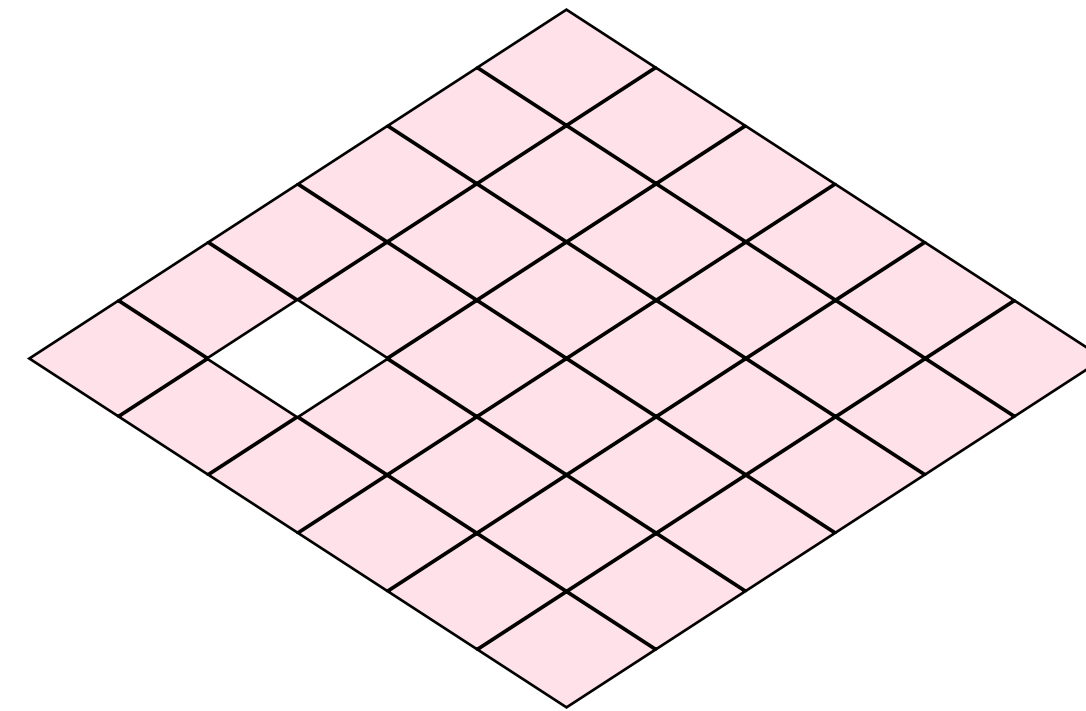
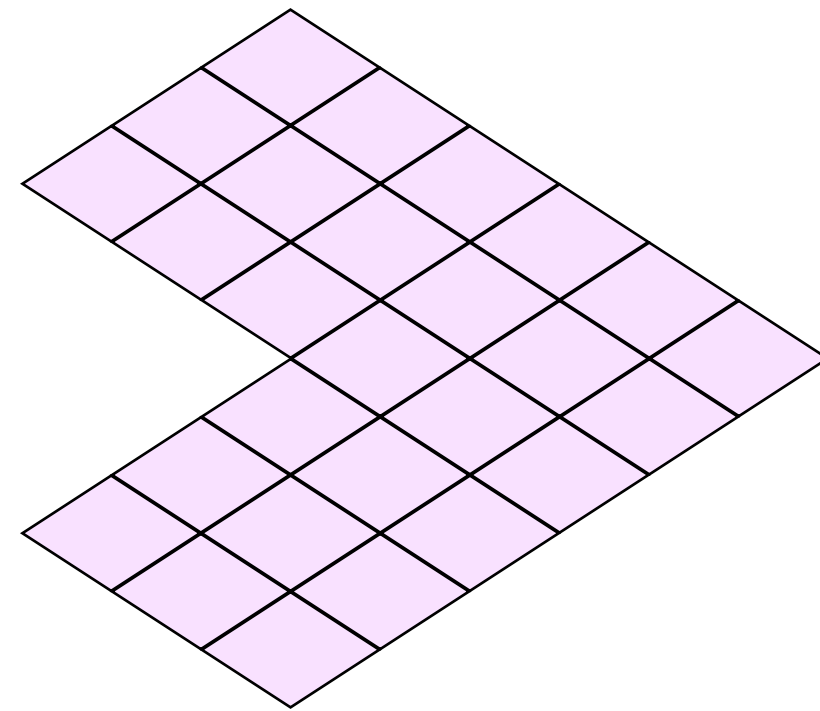
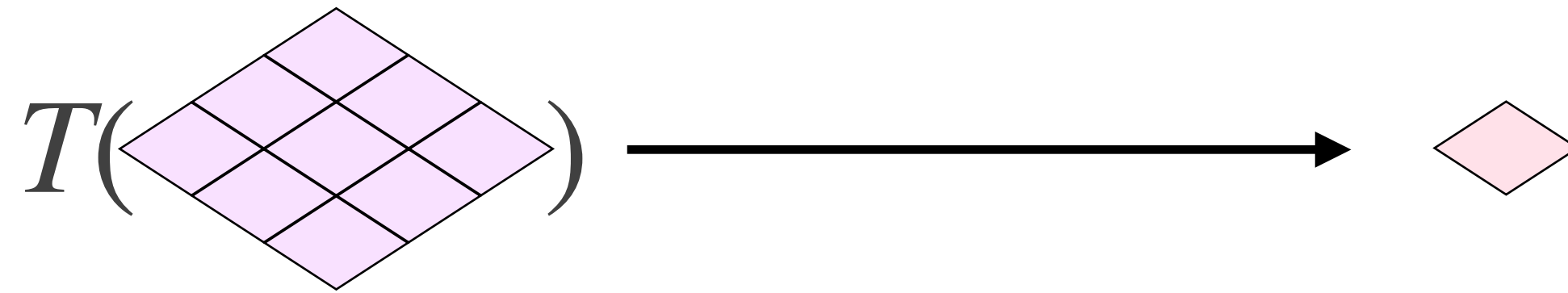
Comparação

E se usássemos uma vizinhança toda para computar uma transformação?

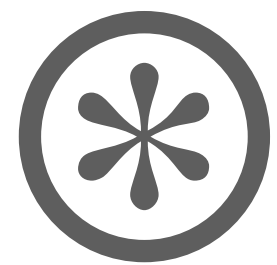


Comparação

E se usássemos uma vizinhança toda para computar uma transformação?



Convolução



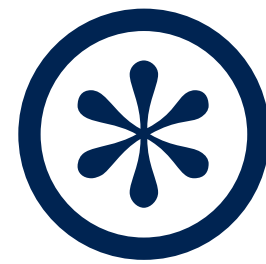
Operação Matemática

Binária

Entre tensores n-dimensionais

Convolução

A



B

a1	a2	a3	a4
----	----	----	----

b1	b2	b3	b4
----	----	----	----

Convolução

A



B

a1	a2	a3	a4
a5	a6	a7	a8
a9	a10	a11	a12

b1	b2	b3	b4
b5	b6	b7	b8
b9	b10	b11	b12

Convolução

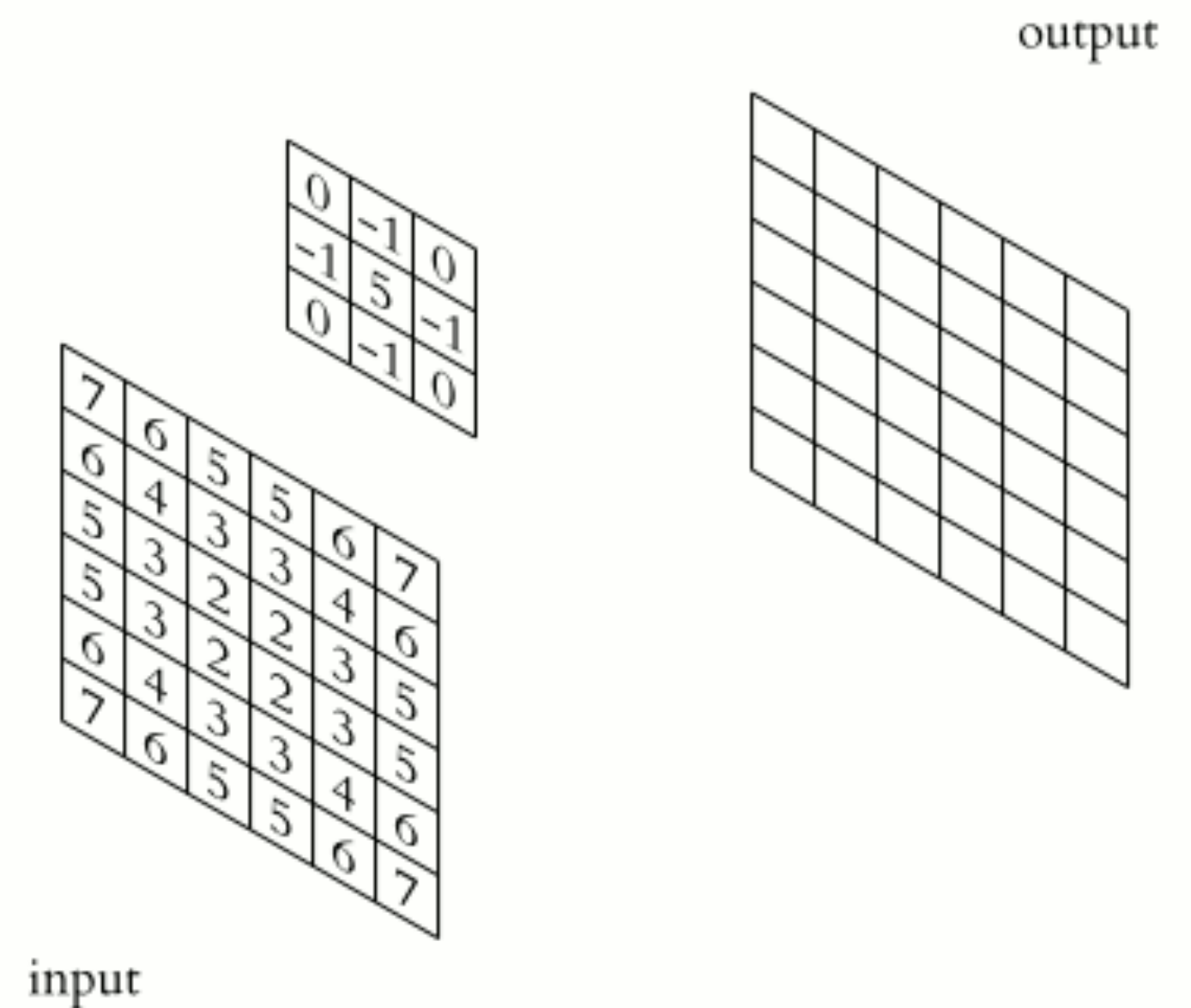
$$C = A \circledast B$$

$$C_{x,y} = \sum_{dx=-a}^a \sum_{dy=-b}^b A_{dx,dy} B_{x-dx,y-dy}$$

Convolução

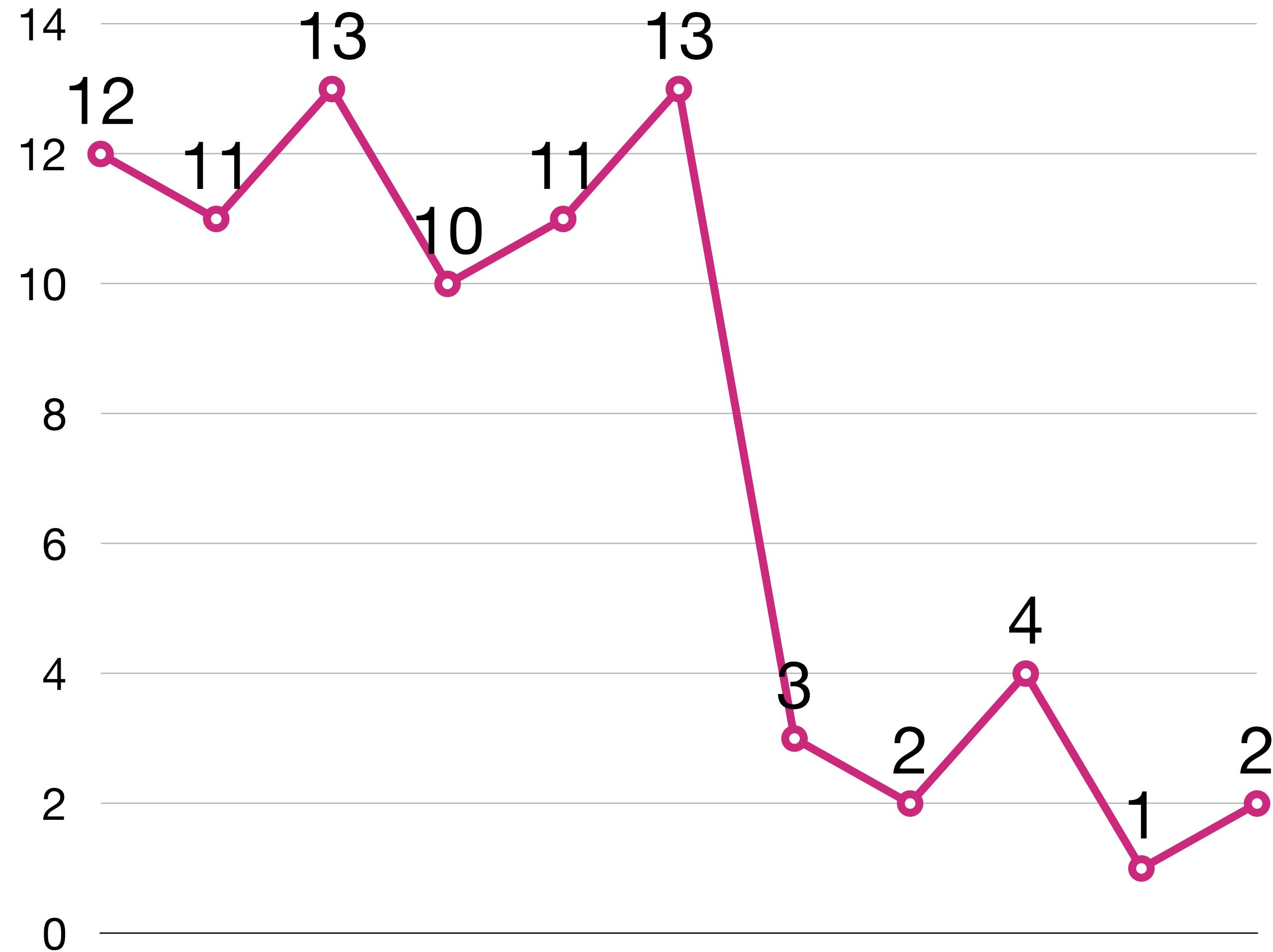
$$C = A \circledast B$$

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Source: Michael Plotke, a wikipedia contributor

Convolução 1D

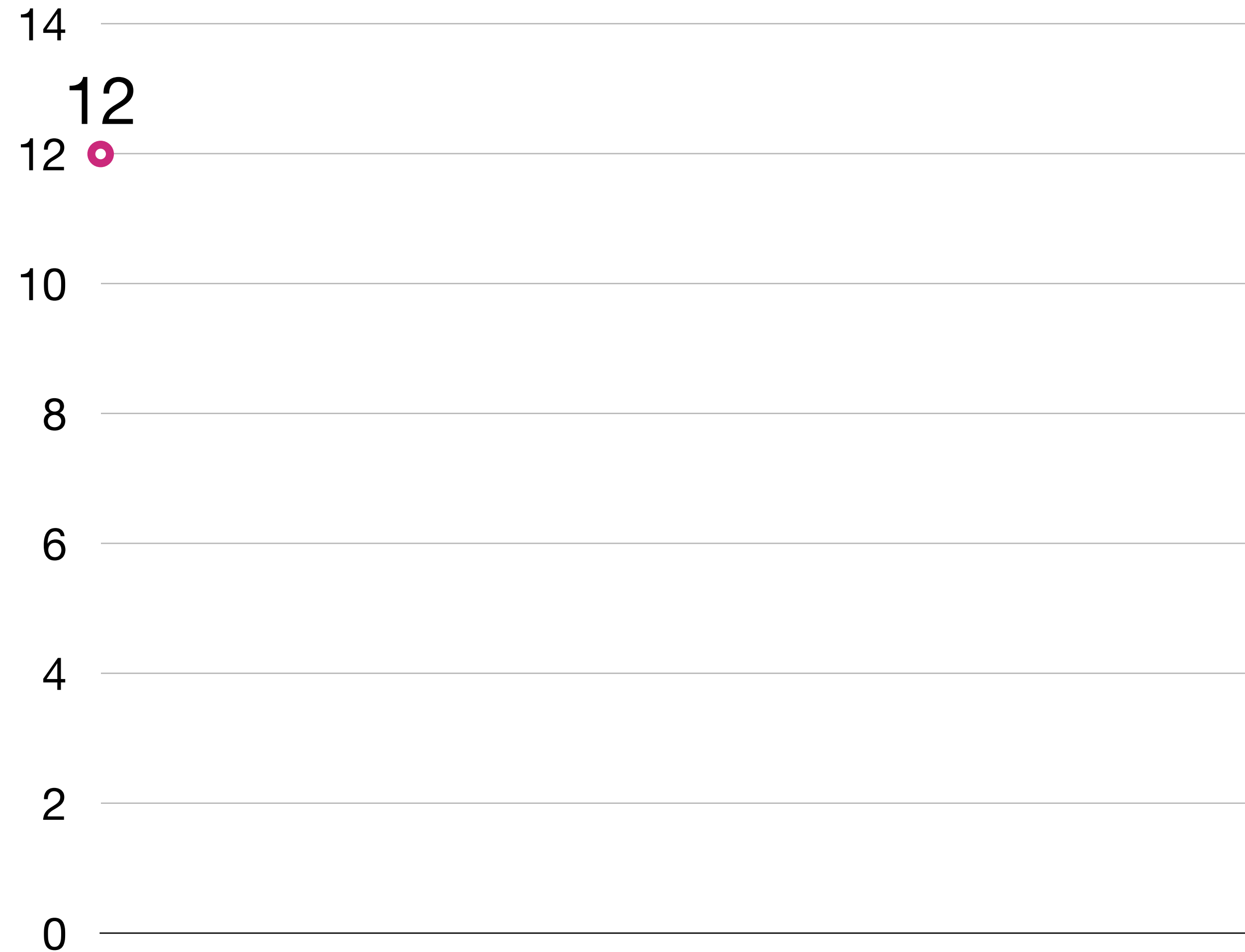


$$C = A \circledast B$$

$$C_x = \sum_{dx=-a}^a A_{dx} B_{x-dx}$$

$$A = \begin{bmatrix} 1/3 & 1/3 & 1/3 \end{bmatrix}$$

Convolução 1D

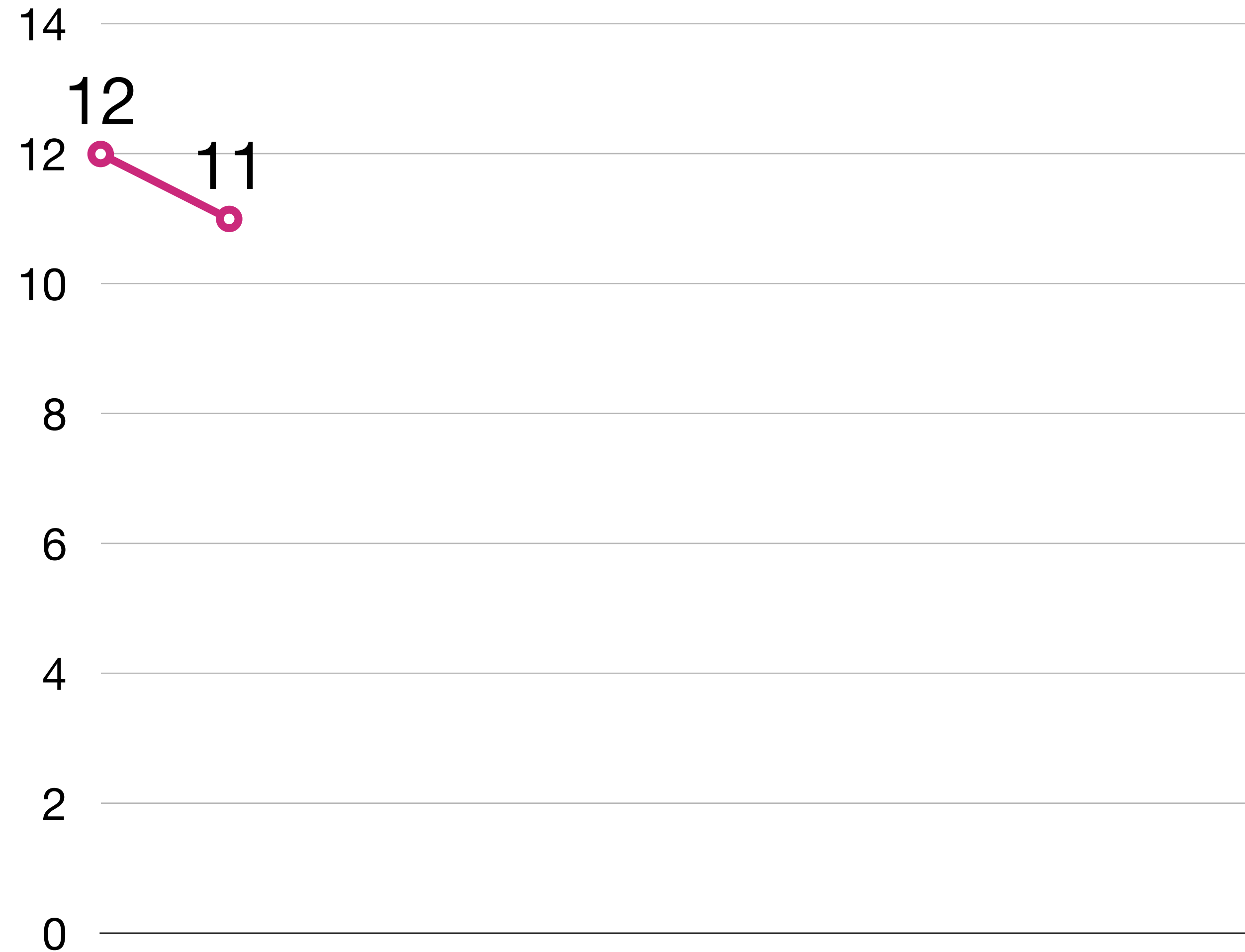


$$C = A \circledast B$$

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Convolução 1D

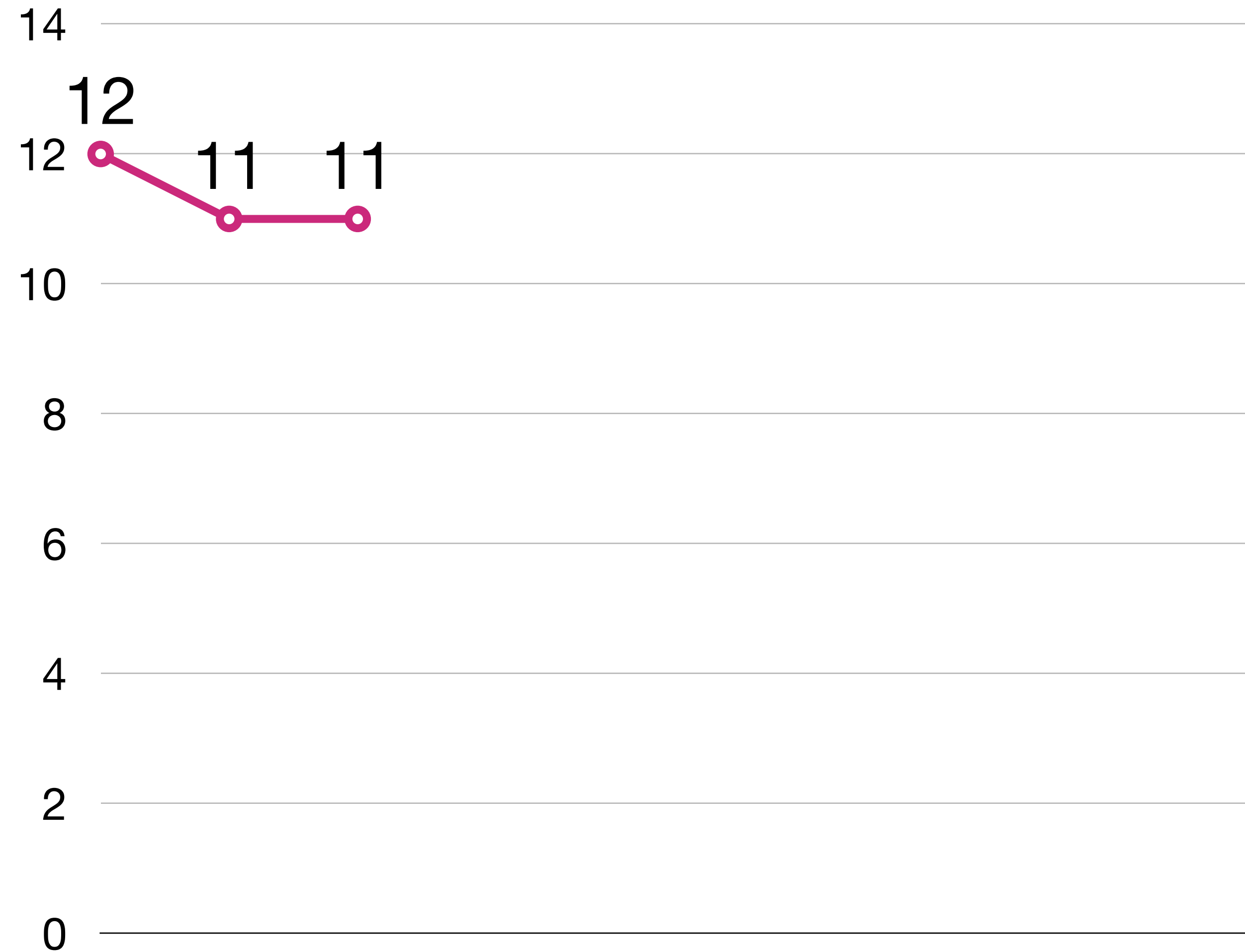


$$C = A \circledast B$$

$$C_x = \sum_{dx=-a}^a A_{dx} B_{x-dx}$$

$$A = \begin{bmatrix} 1/3 & 1/3 & 1/3 \end{bmatrix}$$

Convolução 1D

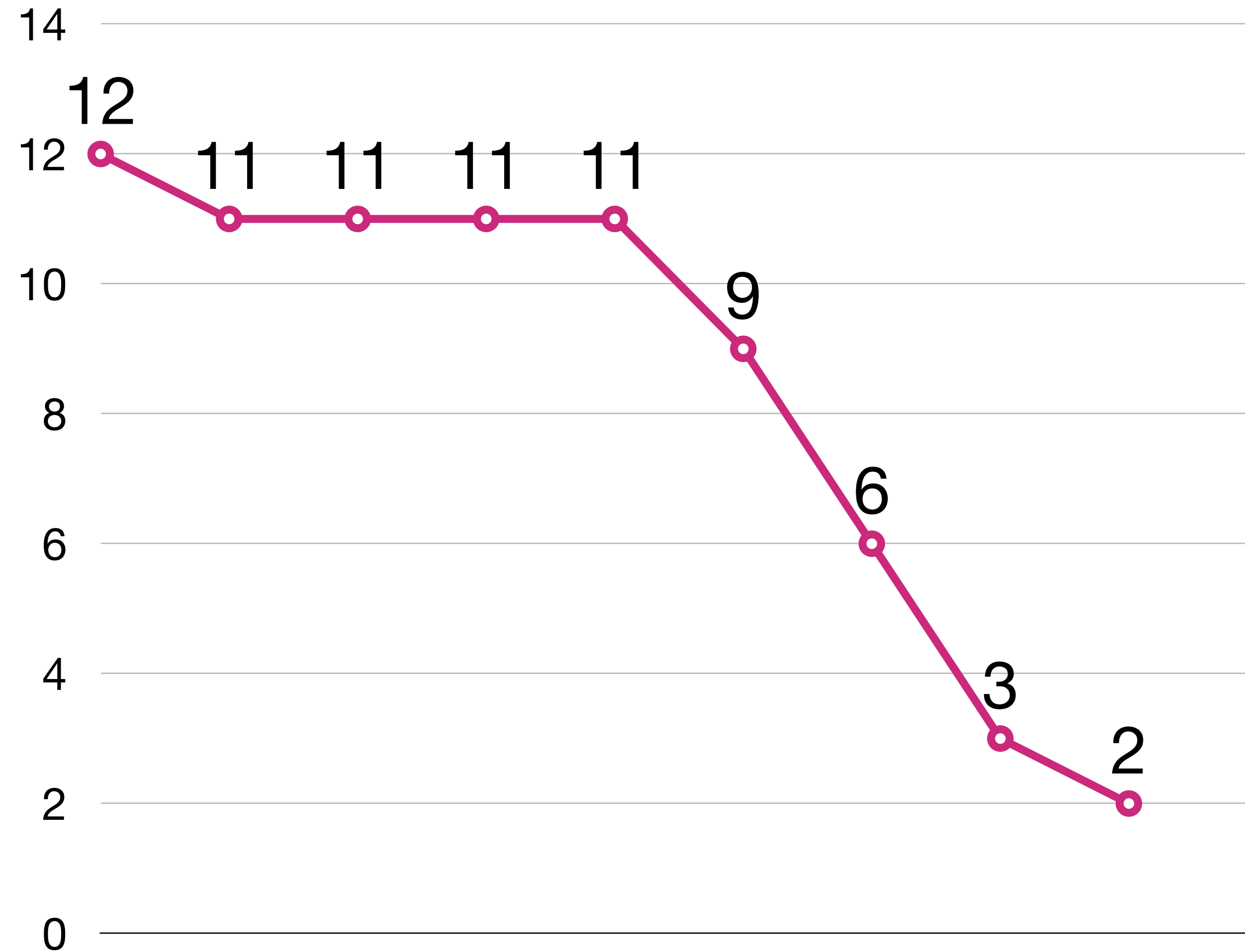


$$C = A \circledast B$$

$$C_x = \sum_{dx=-a}^a A_{dx} B_{x-dx}$$

$$A = \begin{bmatrix} 1/3 & 1/3 & 1/3 \end{bmatrix}$$

Convolução 1D

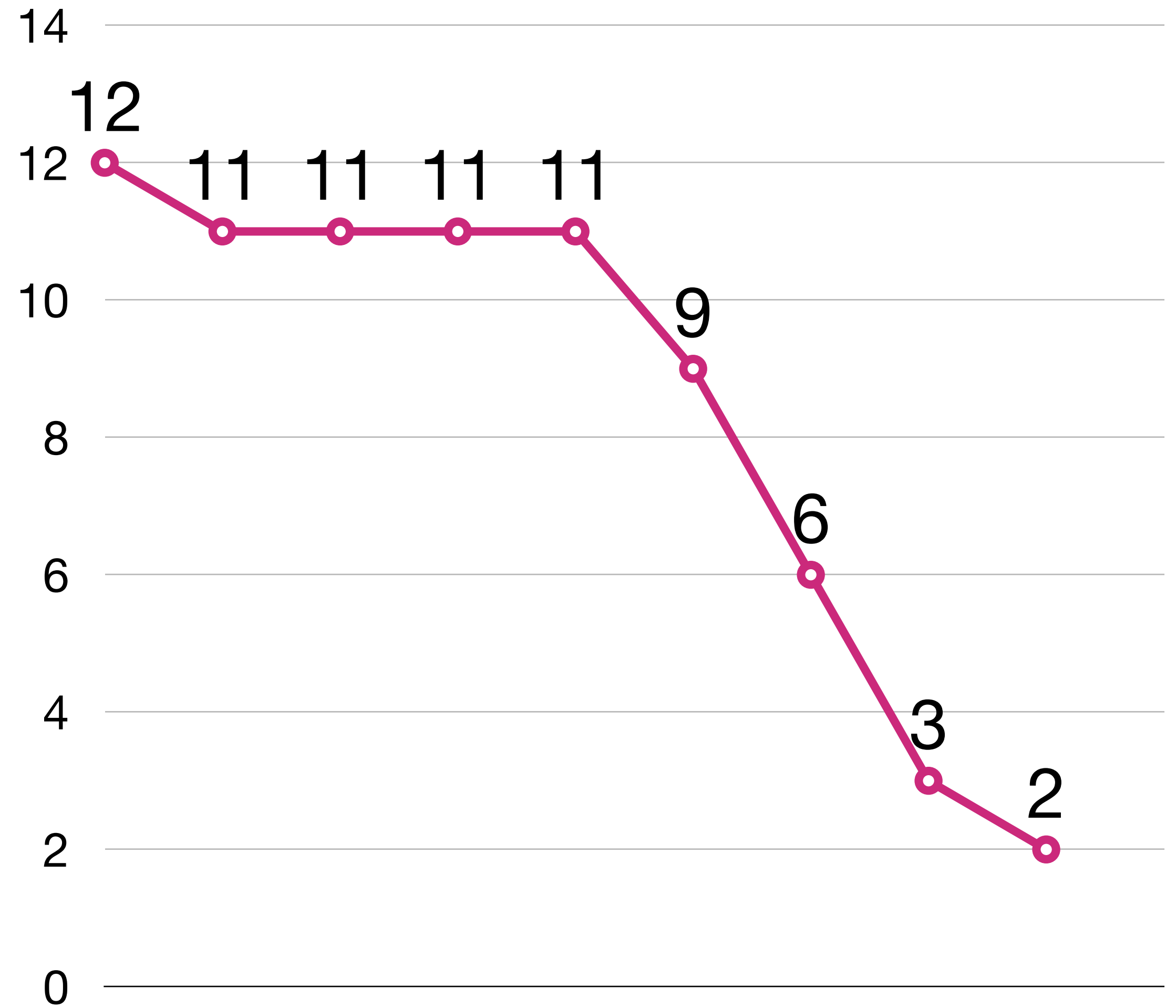
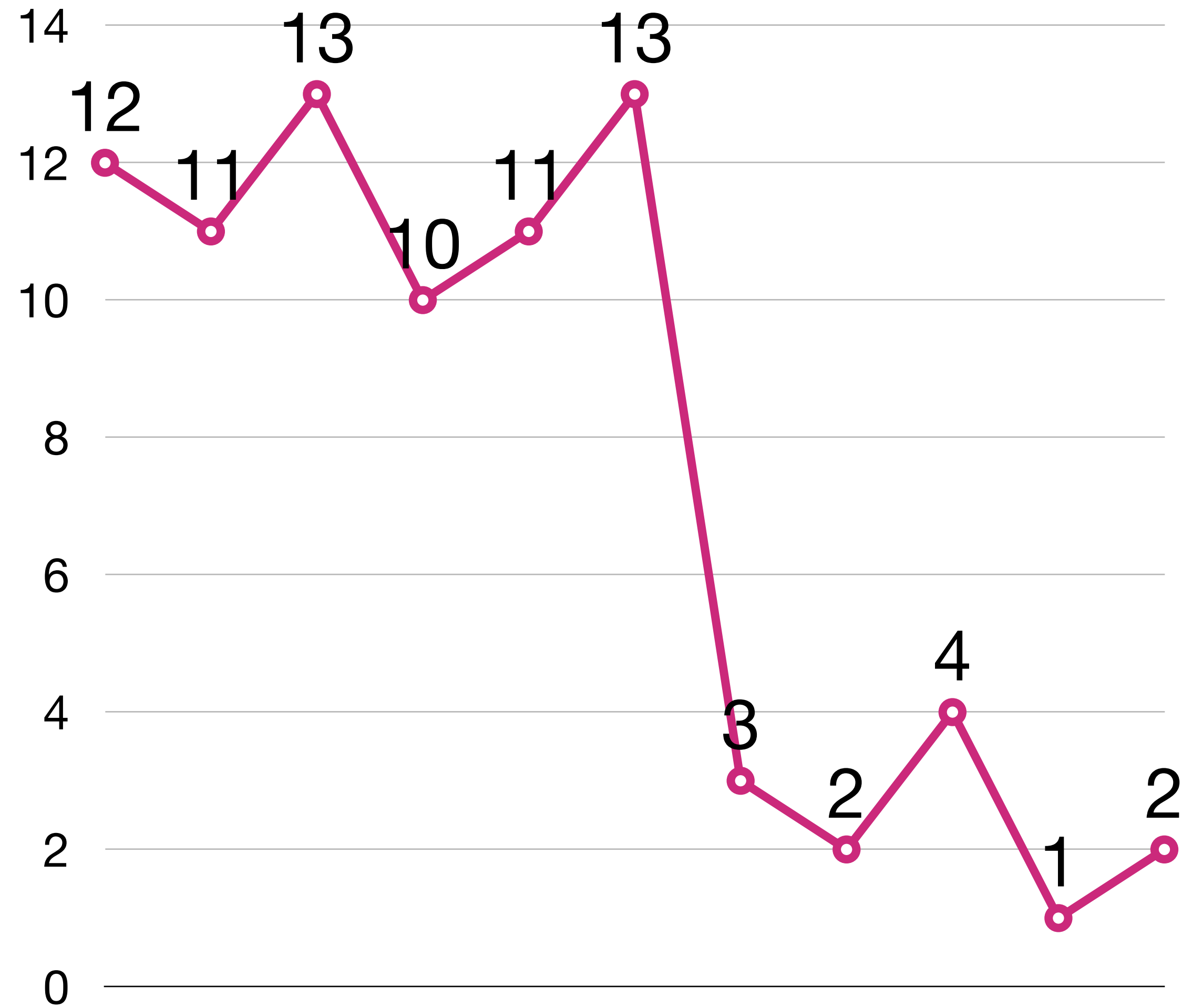


$$C = A \circledast B$$

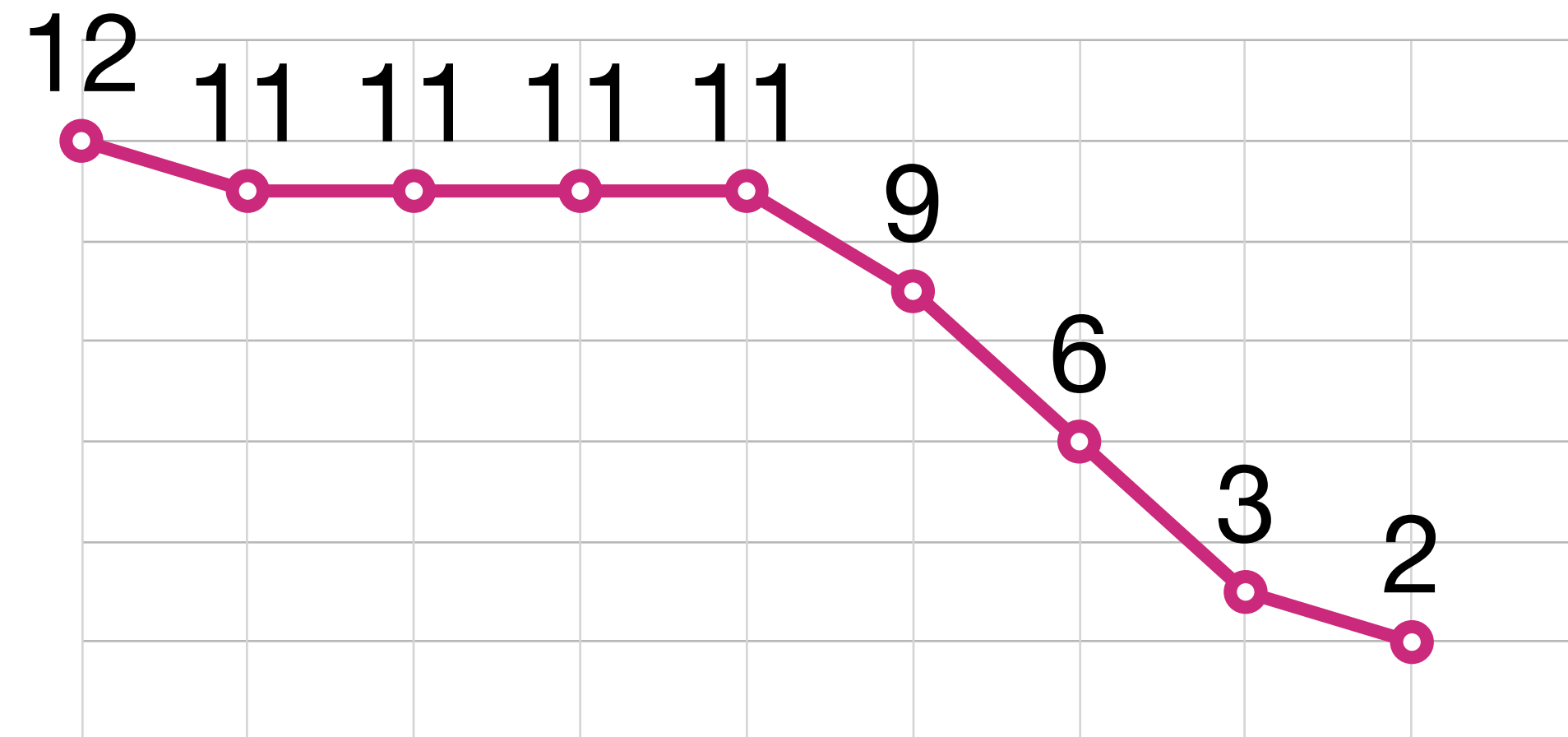
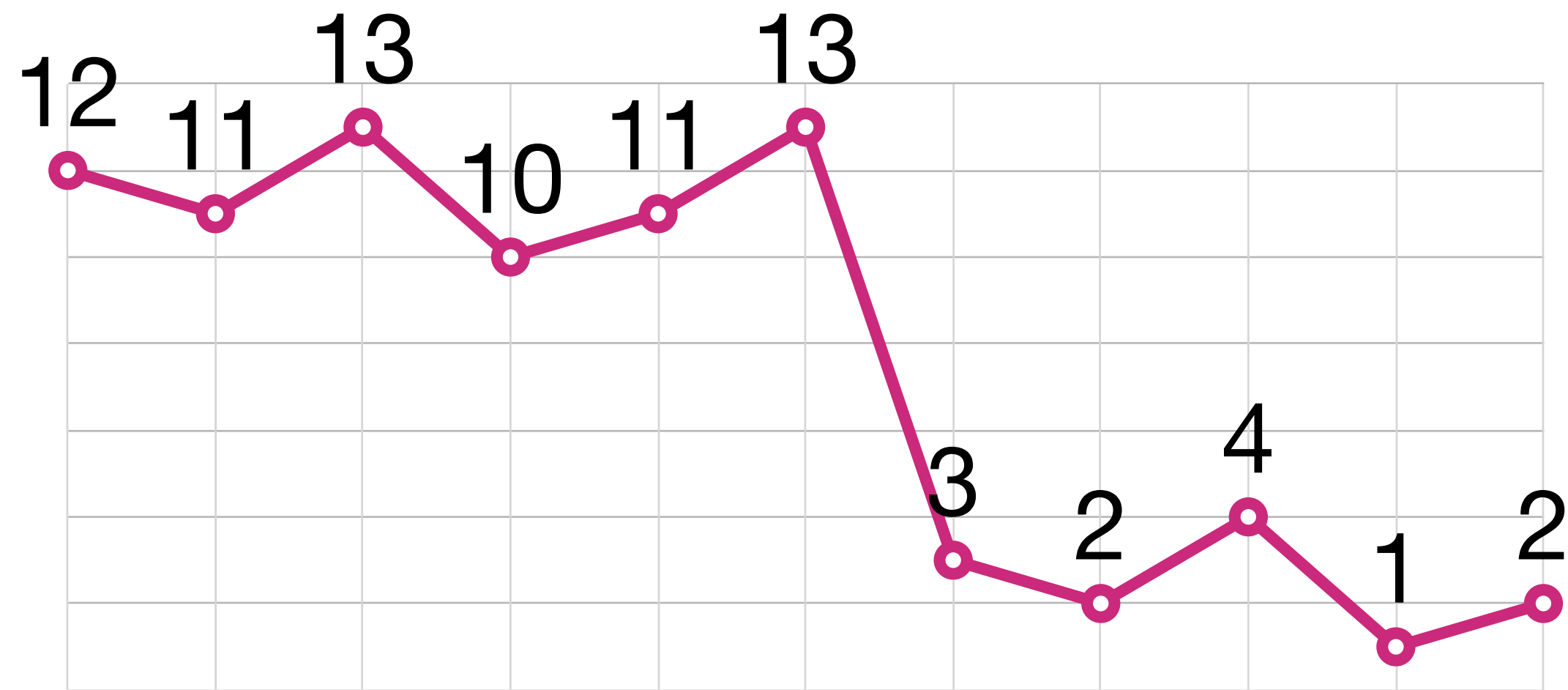
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$$A = \begin{bmatrix} 1/3 & 1/3 & 1/3 \end{bmatrix}$$

Convolução 1D

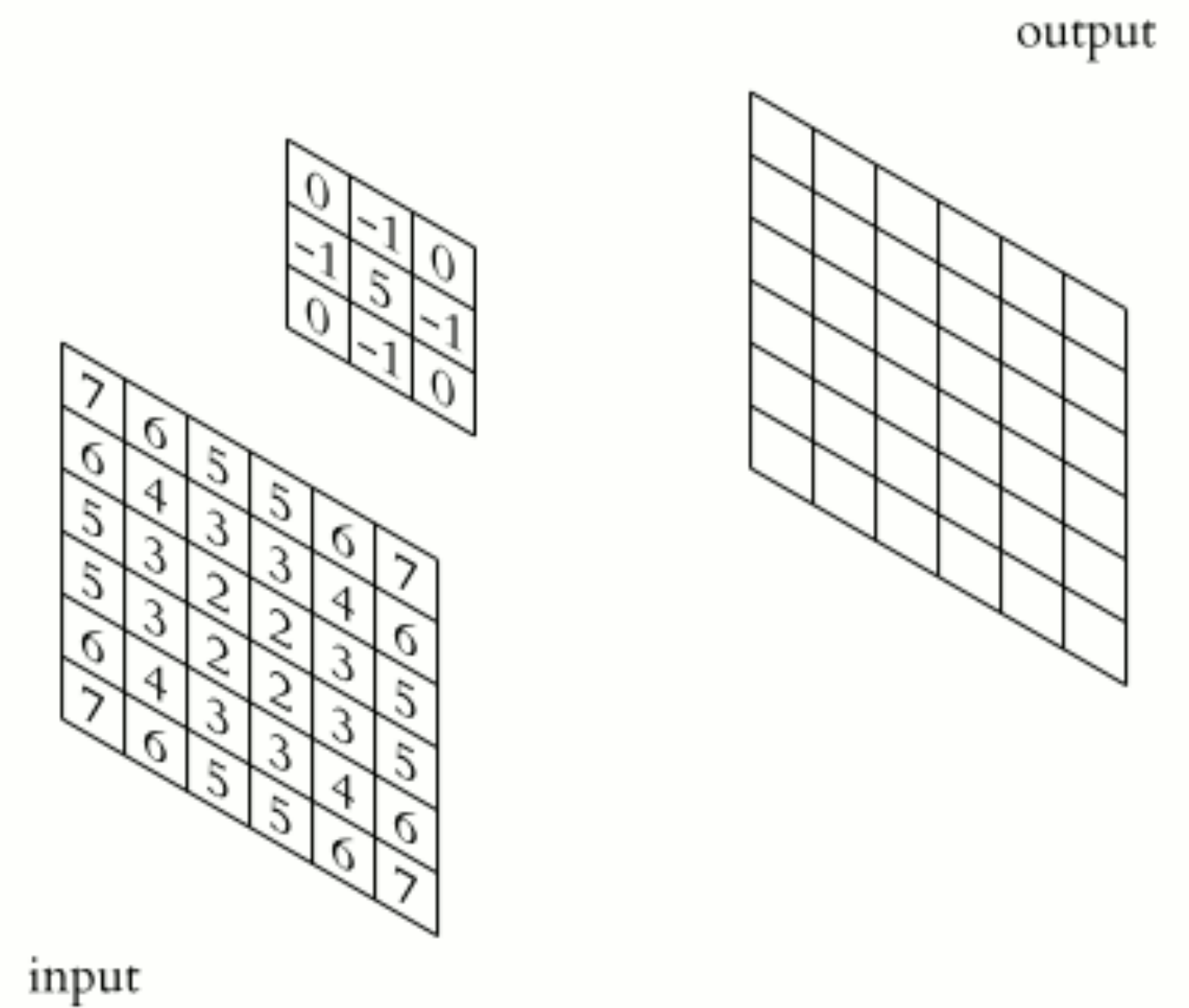


Convolução 1D



Convolução 2D

$$C_{x,y} = \sum_{dx=-a}^a \sum_{dy=-b}^b A_{dx,dy} B_{x-dx,y-dy}$$



Source: Michael Plotke, a wikipedia contributor

Convolução

$$C_{x,y} = \sum_{dx=-a}^a \sum_{dy=-b}^b A_{dx,dy} B_{x-dx,y-dy}$$

entrada

10	5	1	2	10
2	1	0	3	5
5	1	2	10	5
1	2	2	2	1

filtro

-2	1	0
-1	1	0
0	0	2

saída

Convolução

$$C_{x,y} = \sum_{dx=-a}^a \sum_{dy=-b}^b A_{dx,dy} B_{x-dx,y-dy}$$

entrada

10	5	1	2	10
2	1	0	3	5
5	1	2	10	5
1	2	2	2	1

filtro

-2	1	0
-1	1	0
0	0	2

saída

x	x	x	x	x
x				
x				
x				

Convolução

$$C_{x,y} = \sum_{dx=-a}^a \sum_{dy=-b}^b A_{dx,dy} B_{x-dx,y-dy}$$

entrada

	0	1	2	3	
0	10	5	1	2	10
1	2	1	0	3	5
2	5	1	2	10	5
3	1	2	2	2	1

filtro

-2	1	0
-1	1	0
0	0	2

saída

x	x	x	x	x
x				
x				
x				

Convolução

$$C_{x,y} = \sum_{dx=-a}^a \sum_{dy=-b}^b A_{dx,dy} B_{x-dx,y-dy}$$

entrada

	0	1	2	3	
0	10	5	1	2	10
1	2	1	0	3	5
2	5	1	2	10	5
3	1	2	2	2	1

filtro

-2	1	0
-1	1	0
0	0	2

saída

x	x	x	x	x
x				
x				
x				

Convolução

$$C_{x,y} = \sum_{dx=-a}^a \sum_{dy=-b}^b A_{dx,dy} B_{x-dx,y-dy}$$

entrada

	0	1	2	3	
0	10	5	1	2	10
1	2	1	0	3	5
2	5	1	2	10	5
3	1	2	2	2	1

filtro

	-1	0	1
-1	-2	1	0
0	-1	1	0
1	0	0	2

saída

x	x	x	x	x
x				
x				
x				

Convolução

$$C_{x,y} = \sum_{dx=-a}^a \sum_{dy=-b}^b A_{dx,dy} B_{x-dx,y-dy}$$

entrada

	0	1	2	3	
0	10	5	1	2	10
1	2	1	0	3	5
2	5	1	2	10	5
3	1	2	2	2	1

filtro

	-1	0	1
-1	-2	1	0
0	-1	1	0
1	0	0	2

saída

	0	1	2	3	
0	x	x	x	x	x
1	x				
2	x				
3	x				

Convolução

$$C_{x,y} = \sum_{dx=-a}^a \sum_{dy=-b}^b A_{dx,dy} B_{x-dx,y-dy}$$

entrada

	0	1	2	3	
0	10	5	1	2	10
1	2	1	0	3	5
2	5	1	2	10	5
3	1	2	2	2	1

filtro

	-1	0	1
-1	-2	1	0
0	-1	1	0
1	0	0	2

saída

	0	1	2	3
0	x	x	x	x
1	x			
2	x			
3	x			

Convolução

$$C_{x,y} = \sum_{dx=-a}^a \sum_{dy=-b}^b A_{dx,dy} B_{x-dx,y-dy}$$

entrada

	0	1	2	3	
0	10	5	1	2	10
1	2	1	0	3	5
2	5	1	2	10	5
3	1	2	2	2	1

filtro

	-1	0	1
-1	-2	1	0
0	-1	1	0
1	0	0	2

saída

	0	1	2	3
0	x	x	x	x
1	x	19		
2	x			
3	x			

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